

- 2 A fungal pathogen infects melon plants and causes the leaf cells to lose turgor. The melon plant then becomes permanently wilted.

- (a) Suggest a tissue in which the fungal pathogen may be found **and** suggest a reason for your answer.

tissue

reason

[1]

- (b) Breeding experiments between a pure-breeding resistant variety of melon plant, Hemed, and a pure-breeding non-resistant variety, Dulce, were carried out to find out how resistance to the fungal pathogen is inherited.

All the F1 and F2 plants were inoculated with the fungal pathogen.

The numbers of resistant and non-resistant plants were counted 18 days after inoculation with the fungal pathogen.

Fig. 2.1 shows the results of this large-scale breeding programme, involving several thousand melon plants.

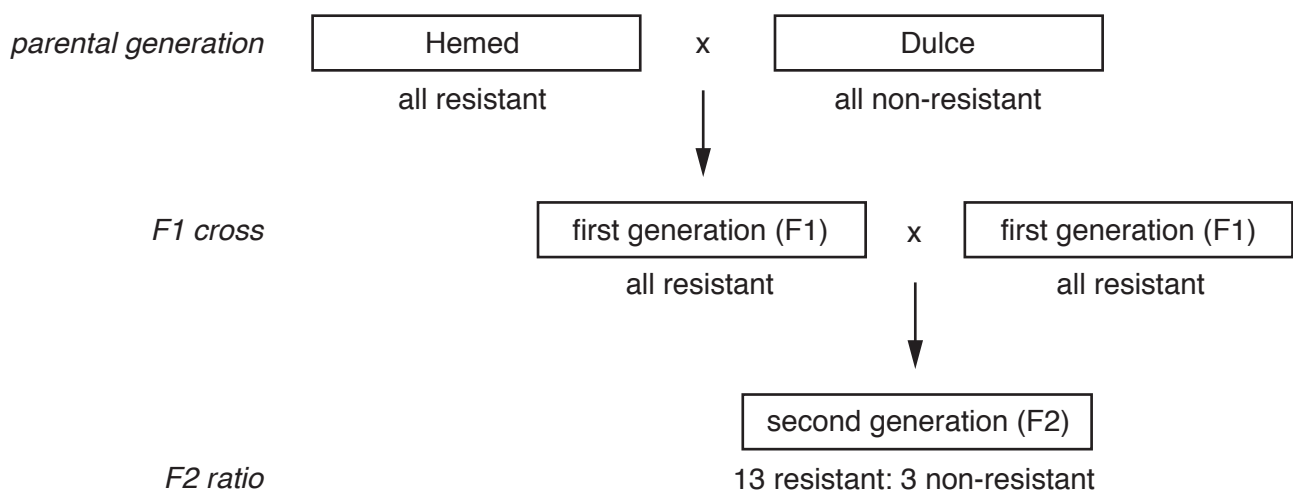


Fig. 2.1

The results in the F2 generation gave a ratio of 13 resistant: 3 non-resistant.

The results in the F2 generation were considered to be anomalous.

Explain why the results for the F2 generation were considered to be anomalous **and** suggest a reason for this ratio.

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..... [2]

(c) Melon plants can be resistant to another fungal pathogen, which has four strains.

Microarrays can be used to study gene expression by detecting mRNA in plant samples.

(i) Outline how microarrays are used to detect mRNA in studies of gene expression in resistant and non-resistant plants.

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..... [3]

Question 2 continues on page 8

Table 2.1 shows the results of a microarray analysis made after the fungal pathogen had infected a resistant variety of melon and a non-resistant variety of melon.

Table 2.1

time after infection with fungal pathogen/hours	number of genes expressed	
	resistant variety	non-resistant variety
24	2461	882
48	821	2237

- (ii) Calculate the ratio of the number of genes expressed at 24 hours for the resistant variety of melon and the non-resistant variety of melon.

Give your answer in whole numbers.

ratio [1]

- (iii) Calculate the percentage change of the genes expressed between 24 hours and 48 hours in the resistant variety.

percentage change = % [1]

- (iv) State what the results in Table 2.1 indicate about resistance to the fungal pathogen.

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..... [2]

[Total: 10]

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